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Magneto-optical conductivity of threefold Hopf semimetals

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Abstract
Multifold Hopf semimetals are a new class of topological materials with intriguing electronic band structures characterized by a dipolar Berry curvature. Using a minimal theoretical model and linear response theory, we theoretically investigate the magneto-optical conductivity of threefold Hopf semimetals in the presence of a strong magnetic field by employing the Kubo formula approach in the zero-temperature limit. Our findings reveal that the optical selection rules vary according to the direction of the applied magnetic field with respect to the dipole axis, resulting in a highly anisotropic magneto-optical conductivity that can be utilized as a distinct spectroscopic signature of threefold Hopf semimetals. In particular, it is shown that some types of optical transitions can be prohibited at certain angles, leading to the disappearance of the corresponding peaks of the magneto-optical conductivity. Our research provides insights into the unique magneto-optical characteristics of threefold Hopf semimetals and their potential use for experimental identification.

Keywords Topological semimetals · Magneto-optical conductivity · Kubo formula

1 Introduction

Topological semimetals, emerging as a novel class of materials, have been widely studied due to their unconventional properties originating from topological band structures unseen in conventional semimetals [1–7]. Weyl semimetals are a prime example of this: the linear band crossing between the conduction band and the valence band serves as a source or sink of Berry curvature, behaving as a single magnetic monopole in the momentum space [5, 6]. Such unusual band and wavefunction structures are well known to cause peculiar phenomena [8–11]. Subsequently, other topological semimetals, such as multi-Weyl semimetals [1], nodal line semimetals [4], chiral multifold semimetals [12, 13], etc., were theoretically predicted and have attracted similarly intensive attention due to their interesting physical properties stemming from the band topological properties.

Recently, a new type of topological semimetals, dubbed as multifold Hopf semimetals, was theoretically proposed [7]. Multifold Hopf semimetals host multiple band crossings, around which the energy band dispersion is linear or flat. Multifold Hopf semimetals are similar to Weyl semimetals in that both have band crossing points that serve as a sink or source of the Berry curvature. The Berry curvature structure of multifold Hopf semimetals, however, acts as a dipolar Berry curvature, which can be regarded as an extended dipole of Weyl semimetals. A dipolar Berry curvature features anisotropic structure, mathematically expressed as

$$\mathbf{\Omega}_\alpha(\boldsymbol{q}) = \kappa_\alpha(\hat{d}\cdot\boldsymbol{q})\frac{\boldsymbol{q}}{|\boldsymbol{q}|^4}, \tag{1}$$

where κ_α is the dipole charge and $\hat{d}$ is the dipole unit vector indicating the dipole direction. This peculiar structure of the dipolar Berry curvature is predicted to strongly affect observables such as anomalous Hall effect, and magneto-conductivities [7]. These observables, however, can also be contributed by trivial Fermi pockets that are not relevant to the dipolar Berry structure, and their use in experimentally discovering multifold Hopf semimetals is limited.

In this paper, we investigate the effects of the dipolar Berry curvature on the magneto-optical conductivity of threefold Hopf semimetals in the presence of a strong magnetic field. We demonstrate that magneto-optical conductivity is highly anisotropic and undergoes intensive qualitative changes as the direction of the applied magnetic field rotates (e.g., disappearance of optical peaks

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at certain critical directions), which can be utilized as a spectroscopic signature of threefold Hopf semimetals.

2 Theory

2.1 Theoretical model for threefold Hopf semimetals

The Hamiltonian of threefold Hopf semimetals in the absence of a magnetic field is given by [7]

$$H^{\xi}(\boldsymbol{q}) = \hbar v \begin{pmatrix} 0 & q_{-}^{\xi} & -iq_z \\ q_{+}^{\xi} & 0 & 0 \\ iq_z & 0 & 0 \end{pmatrix} \tag{2}$$

where $\boldsymbol{q}$ is the momentum measured from the band crossing point, $q_{\pm}^{\xi} = \xi q_x \pm iq_y$ and v is the band velocity. For convenience, we choose the Berry dipole to be aligned with z-axis with the dipole unit vector being $\hat{d} = (0, 0, \xi)$. The sign of ξ indicates that the dipole points toward upward(+) or downward(−) z-axis. The Hamiltonian in the presence of a strong magnetic field can be directly obtained from Eq. (3) by using Peierls substitution, replacing $\boldsymbol{q} \to \boldsymbol{\Pi} = \boldsymbol{q} + e\boldsymbol{A}$:

$$H^{\xi}(\boldsymbol{q}) \to H^{\xi}(\boldsymbol{\Pi}) = \hbar v \begin{pmatrix} 0 & \Pi_{-}^{\xi} & -i\Pi_z \\ \Pi_{+}^{\xi} & 0 & 0 \\ i\Pi_z & 0 & 0 \end{pmatrix} \tag{3}$$

where $\Pi_{\pm} = \xi\Pi_x \pm i\Pi_y$. Without loss of generality, we assume that the magnetic field varies only on the yz-plane, being written as $\boldsymbol{B} = B(0, \sin\theta, \cos\theta)$ with the corresponding magnetic vector potential $\boldsymbol{A} = Bx(0, \cos\theta, \sin\theta)$. Here, θ is the angle of the magnetic field measured from the dipole axis. It is worth noting that the canonical momentum along the direction of the magnetic field (i.e., $q_0 = \hat{\boldsymbol{B}} \cdot \boldsymbol{q}$) should be conserved. The commutation relations between different components of the gauge-invariant moments are calculated to be $[\Pi_x, \Pi_y] = -i\cos\theta/l_B^2$, $[\Pi_y, \Pi_z] = 0$, $[\Pi_z, \Pi_x] = -i\sin\theta/l_B^2$, where $l_B = 1/\sqrt{eB}$ is the magnetic length. It is now easy to see that the ladder operators are defined as [7]

$$\begin{aligned} a &= \frac{l_B}{\sqrt{2}}\left(\Pi_x + i\cos\theta\Pi_y + i\sin\theta\Pi_z\right), \\ a^{\dagger} &= \frac{l_B}{\sqrt{2}}\left(\Pi_x - i\cos\theta\Pi_y - i\sin\theta\Pi_z\right) \end{aligned} \tag{4}$$

obey the standard fermionic commutation relation $[a, a^{\dagger}] = 1$. Using the quantization scheme discussed above, one can obtain the Landau energy levels

$$\varepsilon_{q_0}^{n,\alpha} = \alpha\sqrt{2n + 1 + \xi\cos\theta + q_0^2} \tag{5}$$

and the Landau wavefunctions

$$\Psi_{\alpha}^{n,\xi} = \mathcal{N}\begin{pmatrix} 0 & \varepsilon_{q_0}^{n,\alpha} & 0 \\ \xi - \cos\theta & i\sin\theta q_0 & \xi + \cos\theta \\ \sin\theta & i\cos\theta q_0 & -\sin\theta \end{pmatrix}\begin{pmatrix} \beta_n|n-1\rangle \\ |n\rangle \\ \beta_{n+1}|n+1\rangle \end{pmatrix}, \tag{6}$$

where $\beta_n = \sqrt{eBn/2}$, $\alpha = -1, 0, 1$ is the band index, $n = 0, 1, 2, \ldots$ is the Landau level index with $|n\rangle$ being the number states that the ladder operators act on as $a|n\rangle = \sqrt{n}|n-1\rangle$ and $a^{\dagger}|n\rangle = \sqrt{n+1}|n+1\rangle$. Here $\mathcal{N}$ is the normalization factor that should be calculated after the matrix multiplication.

We plot in Fig. 1 the Landau level energy spectrum at various values of θ as a function of q_0 using the obtained energy dispersion for Landau levels 5. The spectrums are quite similar to those obtained for Weyl semimetals [5], but a noticeable difference is that the band spectrum can be tuned from fully gapped to gapless by simply rotating the applied magnetic field (i.e., by varying θ) because of the presence of the Berry dipole. Another notable feature is that the zero-energy level is heavily degenerate with respect to Landau indices. This plays an important role in determining the optical properties as the zero-level states are involved in the optical transitions, as will be demonstrated in the next section.

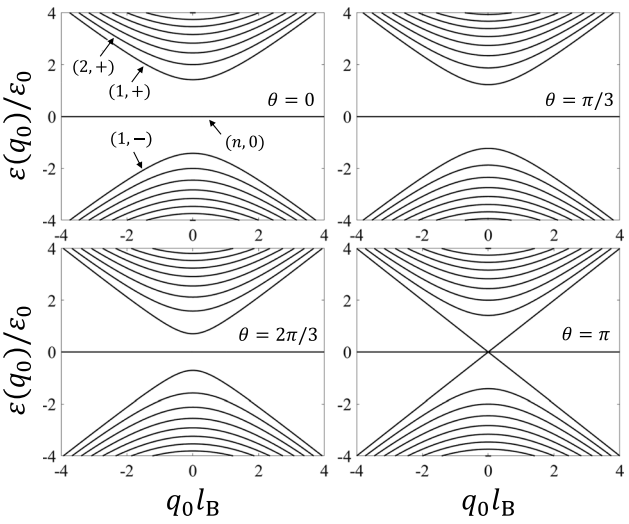


Fig. 1 Plots of Landau energy levels for several values of $\theta = 0, \pi/3, 2\pi/3$ and π, demonstrating a strong dependence of energy band structure on the direction of the applied magnetic field. The labels denoted as (n, α) correspond to a Landau level (n) and band (α) index. Here $\epsilon_0 = \hbar v/l_B$, and we set $\xi = 1$ for calculations

2.2 Magneto-optical conductivity

We employ the Kubo formula approach in the zero-temperature limit to calculate the dynamical magneto-optical conductivity. Expressed in the Landau basis, the Kubo formula is given by

$$\sigma_{ij}(\omega) = -\frac{ie^2}{2\pi l_B^2 \hbar} \sum_{n,m,\alpha,\beta} \int \frac{dq_0}{2\pi} \frac{f_{q_0}^{m,\alpha} - f_{q_0}^{n,\beta}}{\varepsilon_{q_0}^{m,\alpha} - \varepsilon_{q_0}^{n,\beta}} \frac{M_i^{(m,\alpha)(n,\beta)} M_j^{(n,\beta)(m,\alpha)}}{\hbar\omega + \varepsilon_{q_0}^{m,\alpha} - \varepsilon_{q_0}^{n,\beta} + i\eta}. \tag{7}$$

where $i(j) = x, y, z$, $f_{q_0}^{m,\alpha} = \Theta(E_F - \varepsilon_{q_0}^{m,\alpha})$ is the zero-temperature Fermi–Dirac distribution function with Θ denoting the Heaviside step function, $M_i^{(m,\alpha),(n,\beta)} = \langle m, \alpha | V_i | n, \beta \rangle$ is the velocity matrix element and $V_i = \partial H / \partial \Pi_i$ is the velocity operator. Interactions of electrons with themselves or disorders are phenomenologically considered in a single parameter defined as η in the Eq. (7). This phenomenological approach is justified since the interaction effects only lead to a small broadening, without significantly altering the qualitative features. In this work, we only consider the absorption part of the longitudinal optical conductivity (i.e., $\Re[\sigma_{ii}(\omega)]$) for linearly polarized light. In the following calculated results, we omit the symbol $\Re$, which signifies the real part, to simplify the notation.

Optical selection rules play a pivotal role in determining the optical properties of materials from both qualitative and quantitative aspects. In the Kubo formula, optical selection rules are encoded in the velocity matrix element, and thus the analytic form of the velocity matrix element is worth investigating. Evaluating using the obtained Landau level eigenstates (Eq. 6), we have

$$M_x^{(m,\alpha),(n,\beta)} = \frac{1}{A} \begin{cases} \xi\sqrt{\frac{1}{2}}\left[\varepsilon_{q_0}^{n,\beta}\sqrt{m}(\xi - \cos\theta) + \varepsilon_{q_0}^{m,\alpha}\sqrt{n+1}(\xi + \cos\theta)\right], & m = n+1 \\ i\xi q_0 \sin\theta(\varepsilon_{q_0}^{m,\alpha} - \varepsilon_{q_0}^{n,\beta}), & m = n \\ \xi\sqrt{\frac{n}{2}}\left[(\varepsilon_{q_0}^{n,\beta} + \varepsilon_{q_0}^{m,\alpha})\xi + (\varepsilon_{q_0}^{n,\beta} - \varepsilon_{q_0}^{m,\alpha})\cos\theta\right], & m = n-1 \end{cases}$$

$$M_y^{(m,\alpha),(n,\beta)} = \frac{1}{A} \begin{cases} -i\sqrt{\frac{1}{2}}\left[\varepsilon_{q_0}^{n,\beta}\sqrt{m}(\cos\theta - \xi) + \varepsilon_{q_0}^{m,\alpha}\sqrt{n+1}(\xi + \cos\theta)\right], & m = n+1 \\ q_0 \sin\theta(\varepsilon_{q_0}^{m,\alpha} + \varepsilon_{q_0}^{n,\beta}), & m = n \\ i\sqrt{\frac{n}{2}}\left[(\varepsilon_{q_0}^{n,\beta} - \varepsilon_{q_0}^{m,\alpha})\xi + (\varepsilon_{q_0}^{n,\beta} + \varepsilon_{q_0}^{m,\alpha})\cos\theta\right], & m = n-1 \end{cases} \tag{8}$$

$$M_z^{(m,\alpha),(n,\beta)} = \frac{1}{A} \begin{cases} i\sqrt{\frac{1}{2}}\sin\theta\left(\varepsilon_{q_0}^{n,\beta}\sqrt{m} + \varepsilon_{q_0}^{m,\alpha}\sqrt{n+1}\right), & m = n+1 \\ q_0 \cos\theta(\varepsilon_{q_0}^{m,\alpha} + \varepsilon_{q_0}^{n,\beta}), & m = n \\ -i\sqrt{\frac{n}{2}}\sin\theta(\varepsilon_{q_0}^{n,\beta} + \varepsilon_{q_0}^{m,\alpha}), & m = n-1 \end{cases}$$

where $A = 2\sqrt{(1 + 2n + q_0^2 + \xi\cos\theta)(1 + 2m + q_0^2 + \xi\cos\theta)}$. Here and in the following, for simplicity, we express energy and momentum in units of $\varepsilon_0 = \hbar v / l_B$ and $k_0 = l_B^{-1}$. It is important to note that the calculated matrix elements have a strong dependence on θ, which implies that the optical selection rules should vary according to the angle between the applied magnetic field and the dipole axis. For $\theta \neq 0$, the applied magnetic field breaks rotational symmetry along the dipole axis (i.e., dipole axis), and thus σ_{xx} and σ_{yy} should not be equal, as opposed to the case of the non-magnetic field, where the rotational symmetry is fully present and σ_{xx} and σ_{yy} are equal. This can be easily inferred from the evaluated matrix element given by Eq. (9), which shows $\left|M_x^{(m,\alpha),(n,\beta)}\right|^2 \neq \left|M_y^{(m,\alpha),(n,\beta)}\right|^2$ when $\theta \neq 0$. For $\theta = 0$, optical transitions between energy bands with the identical Landau index are allowed(forbidden) for $\sigma_{zz}(\sigma_{xx}$ and $\sigma_{yy})$. The situation becomes exactly reversed as θ increases up to $\pi/2$, implying that the magneto-optical conductivity undergoes a significant qualitative change as θ varies from 0 to $\pi/2$.

In the following section, we present our numerical results along with analytical analysis and demonstrate how the magneto-optical conductivity strongly varies in accordance with the θ-dependent selection rules and thus can be used for experiments aimed at identifying threefold Hopf semimetals.

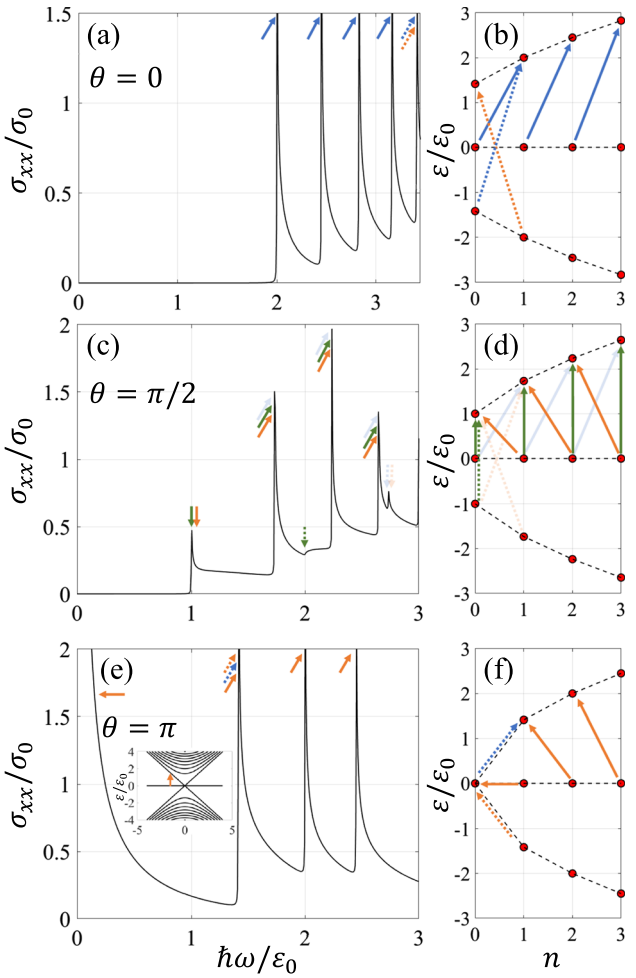


Fig. 2 Real part of calculated optical conductivities (left figures) $\sigma_{xx}(\omega)$ for several values of θ in units of $\sigma_0 = \frac{e^2}{h}k_0$ with $k_0 = l_B^{-1}$ and the snowshoe diagrams representing the transitions corresponding to the optical peaks with colored arrows (right figures). Semitransparent arrows in **c** and **d** correspond to transitions present in **a** and **b**, where the results are plotted at $\theta = 0$. The inset in **e** shows the Landau energy levels, with the orange arrow representing transitions responsible for the strong peak at low frequencies

3 Numerical results

3.1 Intrinsic case ($E_F = 0$)

We first start with the undoped case, where the Fermi energy lies at the zero of energy. For calculations, we shift the Fermi energy upward from zero by an infinitesimal amount, and thus any transitions to flat zero-energy bands are forbidden. Also, without loss of generality, we set $\xi = 1$ throughout the paper for simplicity.

The figures on the left of Fig. 2 present the calculated results for the optical conductivity along the x-axis for various values of $\theta = 0, \pi/2$, and π. We show the corresponding snowshoe diagram in the accompanying figures on the right, where Landau energy levels at $q = 0$ are represented by red dots and plotted as a function of the Landau index n with colored arrows indicating interband transitions corresponding to optical peaks in $\sigma_{xx}(\omega)$. It is worth noting that only transitions $(n, 0) \to (n+1, +)$, $(n+1, -) \to (n, +)$, and $(n, -) \to (n+1, +)$ contribute to optical conductivity. This can be understood from the velocity matrix element calculated at $\theta = 0$, which is written as

$$M_x^{(m,\alpha),(n,\beta)} = \frac{1}{A}\begin{cases} \sqrt{2}\varepsilon_{q_0}^{m,\alpha}\sqrt{n+1}, & m = n+1 \\ 0, & m = n \\ \sqrt{2}\varepsilon_{q_0}^{n,\beta}, & m = n-1 \end{cases} \tag{9}$$

It is immediately clear from Eq. (9) that transitions between the energy bands having the same Landau index are forbidden, i.e., $M_x^{(n,+),(n,0)} = 0$. Furthermore, transitions from zero energy bands with $\alpha = 0$ to higher bands are prohibited in cases where the transitions occur from n to $n-1$ (i.e., $M_x^{(n-1,+),(n,0)} = 0$). As one tunes θ away from 0 to $\pi/2$, the previously forbidden transitions become permissible, as shown in Fig. 2c, d with the two velocity matrix elements at $\pi/2$ given by $M_x^{(n,+),(n,0)} = i\varepsilon_{q_0}^{m,\alpha}q_0$ and $M_x^{(n-1,+),(n,0)} = \sqrt{\frac{n}{2}}\varepsilon_{q_0}^{n-1,+}$. It is worth noting that transitions $(n, -) \to (n, +)$ give rise to a bump-like structure instead of a sharp peak because of a factor of q_0 in $M_x^{(n,+),(n,0)}$, which is not present in other velocity matrix elements. The selection rules undergo significant changes again as one continues to tune θ from $\pi/2$ to π. The previously allowed transitions $(n, 0) \to (n, +)$ and $(n, 0) \to (n+1, +)$ are completely forbidden at $\theta = \pi/2$, leading to a disappearance of the bump structure arising from the transitions $(n, 0) \to (n, +)$. It is also worth noting that there appears a sharp peak at low frequency regime because of interband transitions between the zero-index Landau levels (see the inset of Fig. 2e).

Figure 3a presents the calculated optical conductivity at $\pi/2$, and their corresponding snowshoe diagram is shown in Fig. 3b. At $\theta = 0$ and π, the optical conductivity along the y-axis$[\sigma_{yy}(\omega)]$ is exactly identical to that along the x-axis$[\sigma_{xx}(\omega)]$ because $\left|M_x^{(m,\alpha),(n,\beta)}\right|^2 = \left|M_y^{(m,\alpha),(n,\beta)}\right|^2$, as one can easily see from Eq. (9). Note that the selection rules σ_{yy} are the same as for σ_{xx}, except that transitions $(n, -) \to (n, +)$ are not allowed because $M_y^{(m,-),(n,+)} = 0$ at $\theta = \pi/2$. Thus, σ_{yy} lacks the bump structures in contrast to σ_{xx}, as shown in Fig. 3c.

Figure 4 shows the optical conductivity along the z-axis. Unlike σ_{xx} and σ_{yy}, only transitions from $(n, 0)$ to $(n, +)$ contribute to the optical conductivity at $\theta = 0$, leading to multiple bump structures without sharp optical peaks (Fig. 4a). As is evident from the formula for the velocity matrix element $M_i^{(m,\alpha)(n,\beta)}$ (Eq. 9), optical transitions $n \to n+1$ and $n \to n-1$ become available with increasing

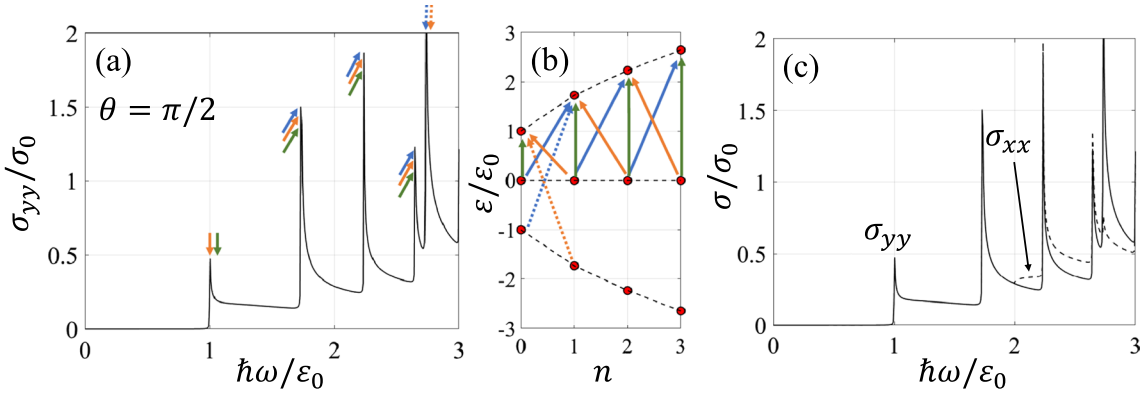


Fig. 3 **a** Calculated optical conductivity along the y-axis and **b** the corresponding snowshoe diagram. **c** Optical conductivity along the x- and y-axes at $\theta = \pi/2$, each of which is plotted as dotted and solid lines, respectively

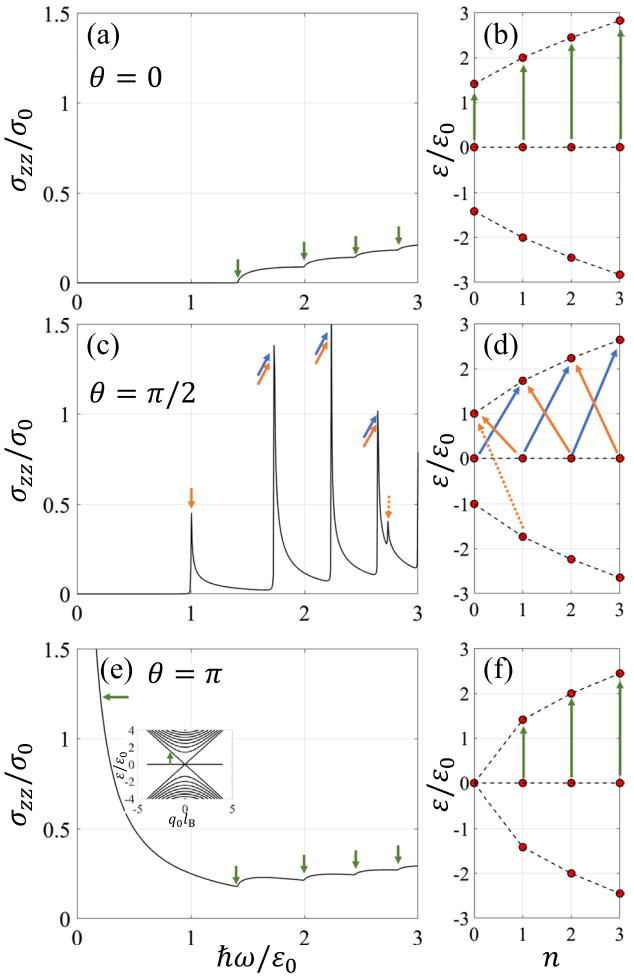


Fig. 4 Real part of optical conductivities along the z-axis (left figures) and the corresponding snowshoe diagram (right figures). The notation is the same as Fig. 2

θ away from 0, while the previously allowed transitions at $\theta = 0$, i.e., $n \to n$ become weaker. This leads to sharp optical peaks in σ_{zz} at $\theta = \pi/2$, as shown in Fig. 4c. The selection rules at $\theta = \pi$ is exactly the same as that at $\theta = 0$, and thus σ_{zz} at $\theta = \pi$ looks quite similar to σ_{zz} at $\theta = 0$. The only noticeable difference is that a wide peak is present at low frequencies when $\theta = \pi$ because of the gapless energy band structure of the zero Landau energy levels.

In Fig. 5, we present comprehensive density plots of optical conductivity, illustrating the evolution of optical conductivity with respect to θ. It is important to note that not only the presence but also the position of the optical peaks varies significantly with θ, which is the main result of this paper.

Our discussion has been so far limited to an intrinsic case where the Fermi level is ideally located at zero-energy. In the following section, we consider a more realistic situation where the system is doped up to a finite chemical potential and demonstrate how it impacts the optical conductivity.

3.2 Extrinsic case ($E_F \neq 0$)

3.2.1 Intraband transitions

When the system is doped with the Fermi energy away from the neutral point, the mobile electrons at the Fermi energy contribute to the optical conductivity, forming a Lorentzian Drude peak in the optical spectrum at low frequencies:

$$\sigma(\omega) \approx D(E_F)\frac{\eta}{\pi(\omega^2+\eta^2)}. \tag{10}$$

Here $D(E_F)$ is the Drude weight. In the Kuba formula we present in Eq. (7), this situation corresponds to intraband transitions [i.e., transitions between the same band $(n,\alpha) \to (n,\alpha)$]. The corresponding velocity matrix elements are calculated to be

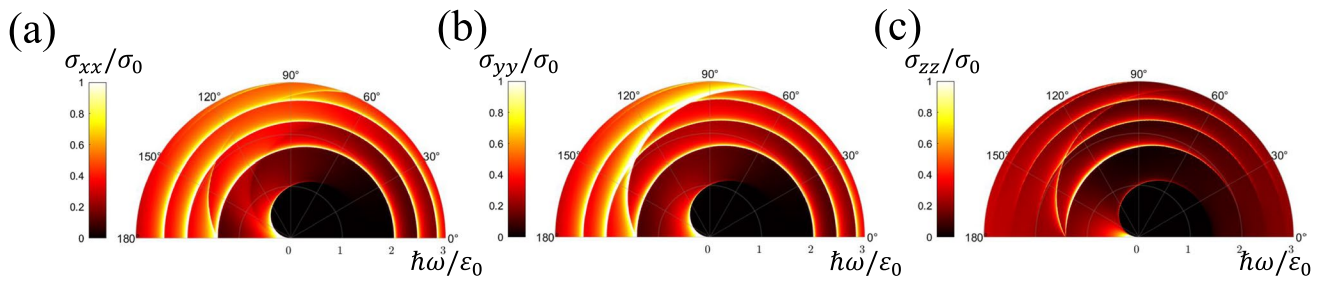


Fig. 5 Two-dimensional plots of optical conductivity along **a** x-, **b** y-, and **c** z-axes. The optical conductivity is represented by the color bar with darker colors corresponding to lower values. The radial direction represents the optical energy in units of ε_0 and the angular direction represents the angle between the applied magnetic field and the dipole axis

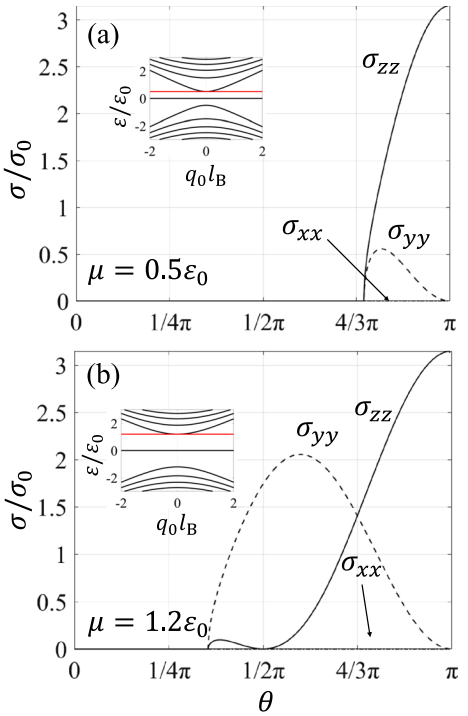


Fig. 6 Optical conductivity at a fixed low-frequency ($\omega = 0.01\varepsilon_0$) with dot-dashed, dashed and solid lines representing σ_{xx}, σ_{yy} and σ_{zz}, respectively. The insets show the Landau level spectrum at the values of θ corresponding to the onset of the intraband optical conductivity. Here we set $\eta = 0.001\varepsilon_0$

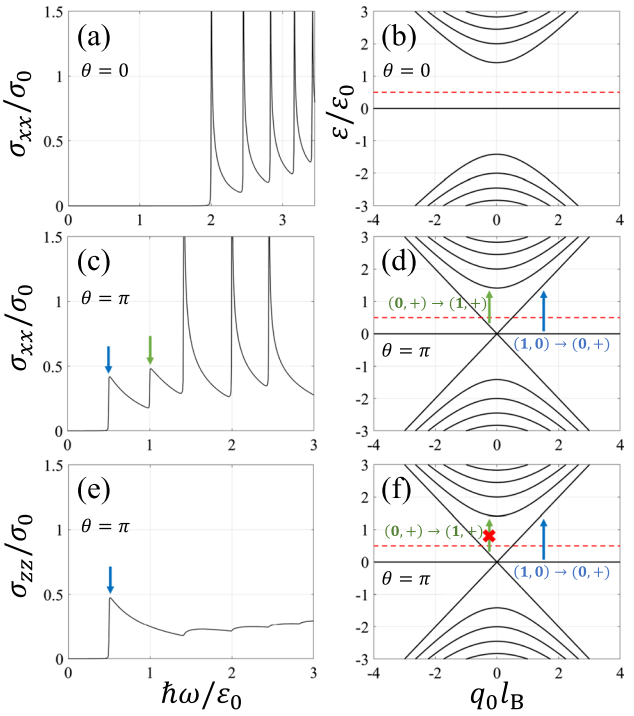


Fig. 7 Real part of optical conductivities along the x-axis (left figures) for various values of θ and the corresponding Landau level spectrums (right figures). Here red dashed lines and arrows in the right figures respectively indicate the Fermi energy and the interband transitions responsible for the optical peaks in the left figures

$$
\begin{aligned}
M_x^{(n,\alpha),(n,\alpha)} &= 0, \\
M_y^{(n,\alpha),(n,\alpha)} &= \frac{2}{A} q_0 \sin\theta \varepsilon_{q_0}^{n,\alpha}, \\
M_z^{(n,\alpha),(n,\alpha)} &= \frac{2}{A} q_0 \cos\theta \varepsilon_{q_0}^{n,\alpha}.
\end{aligned}
\tag{11}
$$

It is important to note from Eq. (11) that the intraband transitions are completely forbidden due to selection rules when the absorbed light is linearly polarized along the x-axis. For lights polarized along different directions, the selection rules vary strongly with θ, implying that the selection rule changes according to the direction of the applied magnetic field. At $\theta = 0$, the selection rules allow intraband transitions only when the incident light is polarized along the z-axis. In contrast, at $\theta = \pi/2$, intraband transitions are allowed only when the light is polarized along the y-axis. These findings imply that, even within the low-frequency regime, low-frequency optical conductivity can help to discover multi-Hopf semimetals by monitoring the evolution of the Drude conductivity as the direction of the applied magnetic field changes.

Figure 6 shows the optical conductivity along the three Cartesian axes calculated at a fixed low-frequency $\omega = 0.01\varepsilon_0$ where the intraband (i.e., Drude) conductivity appears. The intraband optical conductivity is zero at small values of θ because there are no conducting channels with μ in the energy gap. Intraband transitions are available at critical values of θ where the chemical potential crosses the upper part of the energy band gap (see insets of Fig. 6). Thus, the optical conductivity of Drude appears only at $\theta > \theta_c$ where θ_c denotes the critical values of θ. Note that Drude conductivity as a function of θ follows the selection rules explained in the previous paragraph: For σ_{xx}, the optical conductivity should always be zero, while σ_{yy} and σ_{zz} are zero at $\theta = \pi$ and $\theta = \pi/2$, respectively. This result shows that Drude conductivity has a very strong dependence on the direction of the magnetic field.

3.2.2 Interband transitions

Figure 7 presents the calculated optical conductivities at $\theta = 0$ and $\theta = \pi$ in the presence of only interband transitions with the intraband transitions neglected for brevity. Note that the results for $\theta = 0$ are the same as those obtained when the Fermi level is at zero energy because both the cases have the same interband transition channels. However, at $\theta > \theta_c$, where θ_c is the critical angle as defined in the previous subsection, the lowest Landau energy level ($(0,+)$) is partially occupied, allowing for a new optical transition $(0,+) \to (1,+)$, as shown in Fig. 7. To investigate the selection rules, it is worth looking into the velocity matrix elements for $(n-1,+) \to (n,+)$:

$$
\begin{aligned}
M_x^{(n-1,+),(n,+)} &= \frac{\sqrt{2}}{A}\left[\varepsilon_{q_0}^{n,+}\cos^2\left(\frac{\theta}{2}\right) + \varepsilon_{q_0}^{n-1,+}\sin^2\left(\frac{\theta}{2}\right)\right],\\
M_y^{(n-1,+),(n,+)} &= \frac{\sqrt{2}i}{A}\left[\varepsilon_{q_0}^{n,+}\cos^2\left(\frac{\theta}{2}\right) - \varepsilon_{q_0}^{n-1,+}\sin^2\left(\frac{\theta}{2}\right)\right]\\
M_z^{(n-1,+),(n,+)} &= -\frac{i}{A}\sqrt{\frac{n}{2}}\sin\theta(\varepsilon_{q_0}^{n,+} + \varepsilon_{q_0}^{n-1,+}).
\end{aligned}
\tag{12}
$$

One can easily see from Eq. (12) that the interband transitions $(0,+) \to (1,+)$ is forbidden for σ_{zz} at $\theta = 0$ and π, exhibiting a strong anisotropy. This can be numerically verified from Fig. 7e, f, where the optical peak of σ_{zz} corresponding to the optical transition $(0,+) \to (1,+)$ is absent, in contrast to σ_{xx} shown in Fig. 7c. Figure 8 presents a two-dimensional plot of optical conductivity in the presence of a finite chemical potential, summarizing the results of this section. Note that the optical conductivities exhibit strong anisotropy similar to the intrinsic case discussed in the previous section.

4 Summary

We have theoretically investigated the optical response of threefold Hopf semimetals in the presence of a strong magnetic field. Our detailed analysis of the selection rules based on the Kubo Formula (Eq. 7) show that the selection rules are distinct for different angles between the applied magnetic field and the dipole axis. In particular, it is shown that some optical transitions can be prohibited at certain angles, leading to a disappearance of optical peaks. Drude peaks are also subject to angle-dependent selection rules, implying that the optical conductivity should be anisotropic within a low-frequency regime comparable to the disorder energy scale. In conclusion, our analytical and numerical results show that the optical conductivity of threefold Hopf semimetals is strongly anisotropy due to the presence of the Berry dipole axis, which can be utilized as a spectroscopic signature of threefold Hopf semimetals.

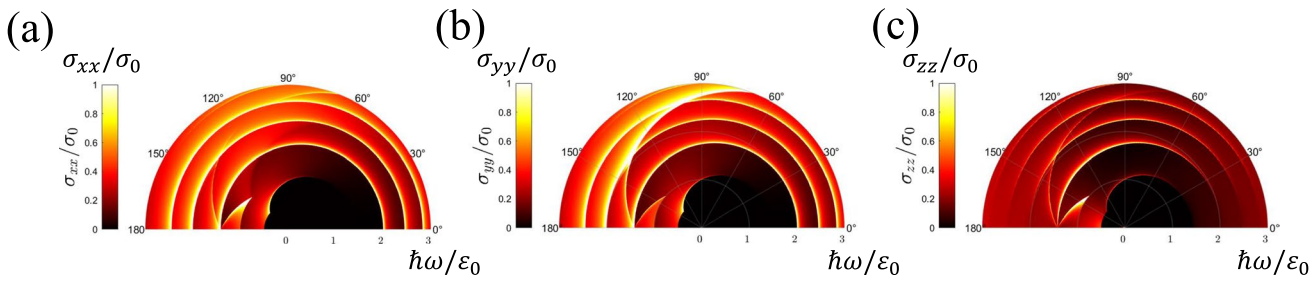


Fig. 8 Same as Fig. 5 except that the chemical potential $\mu = 0.5\varepsilon_0$ is finite here

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